When Random Noise Looks Predictive

A reproducible demonstration of data leakage in machine learning

Project type: Methodological demonstration using synthetic data

Tools: Python, NumPy, pandas, scikit-learn, Matplotlib, Google Colab

1. Objective

This project investigates how supervised feature selection performed before cross-validation can leak label information into validation folds and produce an overly optimistic estimate of model performance. The purpose is to demonstrate a machine-learning evaluation pitfall, not to build a clinical prediction tool.

2. Methods

For each of 10 random seeds (0–9), a synthetic dataset was generated with 200 observations, 5,000 normally distributed random numerical features, and random binary labels independent of all features. Logistic regression was evaluated using five-fold stratified cross-validation.

Flawed method: SelectKBest with an ANOVA F-test selected the 100 features with the strongest apparent association with the labels using the full dataset before cross-validation. This allows validation-fold labels to influence feature selection.

Corrected method: Feature selection, standardization, and logistic regression were placed in a single scikit-learn pipeline. Each fold therefore fits feature selection only on that fold's training partition.

3. Results

Seed	Flawed AUC	Correct AUC
0	0.9960	0.5752
1	0.9409	0.5193
2	0.9534	0.4727
3	0.9865	0.4737
4	0.9820	0.5338
5	0.9654	0.4568
6	0.9760	0.4634
7	0.9920	0.5229
8	0.9840	0.5135
9	0.9834	0.6002
Mean	**0.9760**	**0.5131**

The mean ROC-AUC was 0.9760 for the flawed method and 0.5131 for the corrected method. The random-ranking reference is 0.5000. The gap between the two observed means is 0.4628 AUC points.

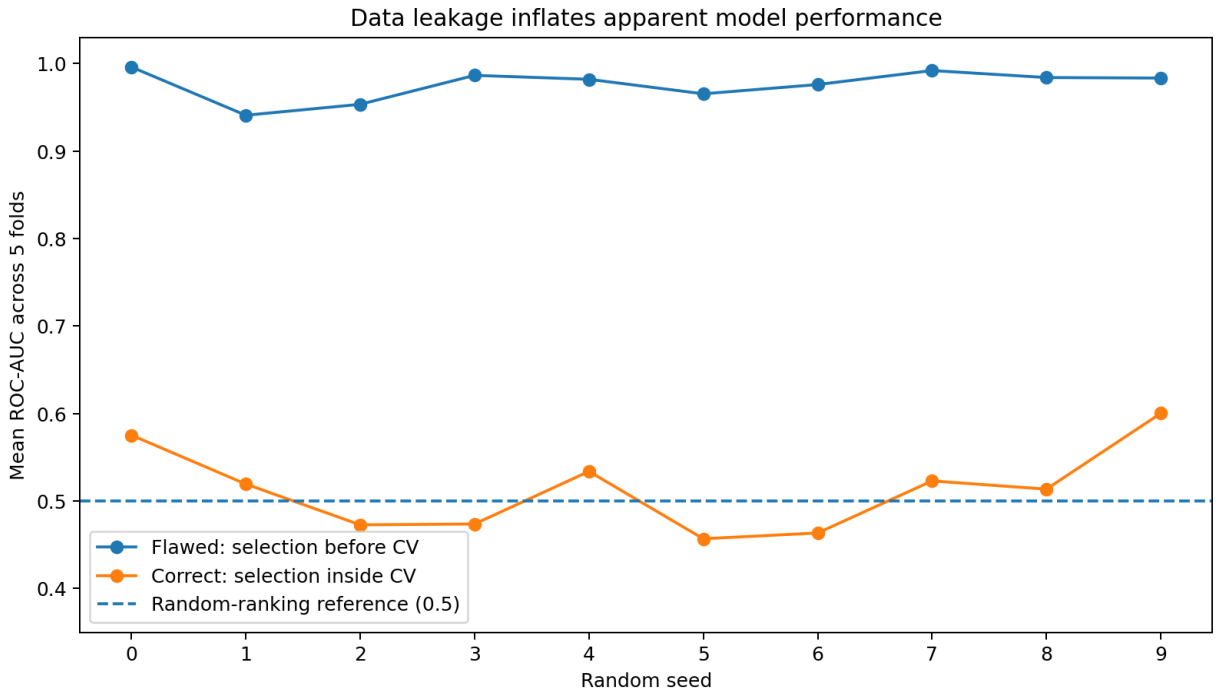


Figure 1. Mean five-fold ROC-AUC by random seed. The flawed method is consistently inflated in these observed runs.

4. Interpretation

The flawed procedure used all outcome labels to choose features, including labels from samples later treated as validation data. The validation process was therefore not independent. The corrected pipeline prevented this particular leakage by fitting feature selection separately within each training fold. Its mean score of 0.5132 was much closer to the 0.5 random-ranking reference, as expected when features contain no true predictive signal.

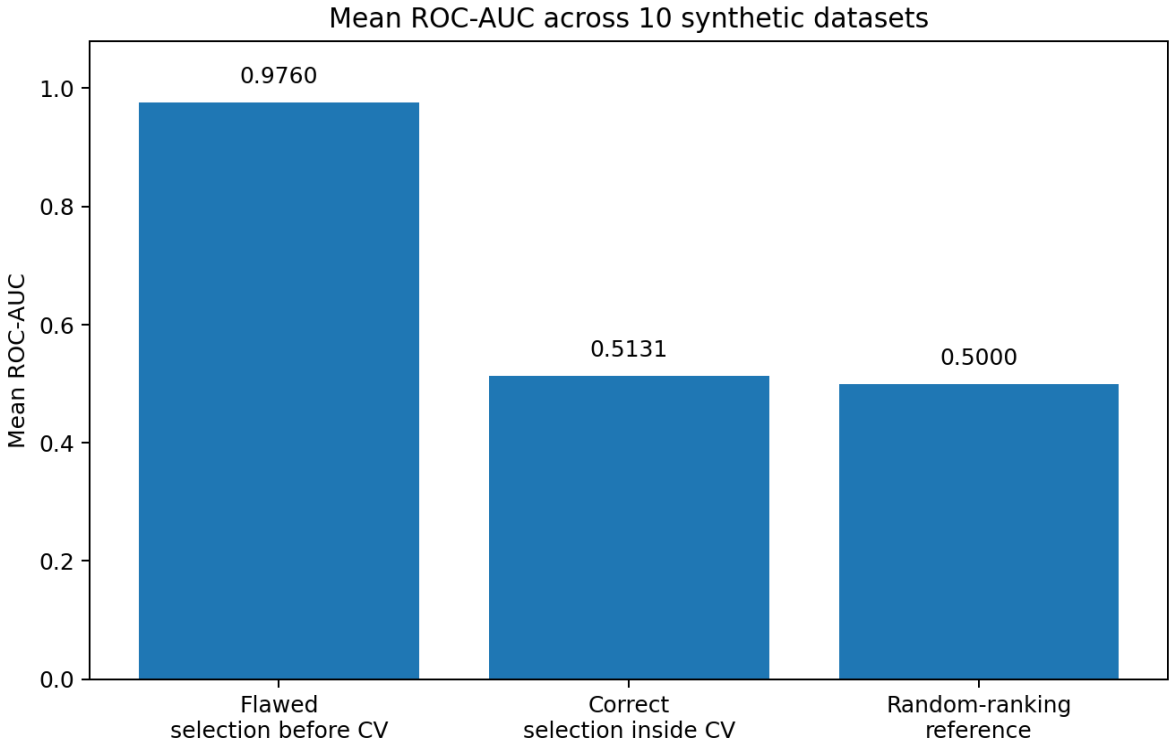


Figure 2. Mean ROC-AUC comparison across ten synthetic datasets.

5. Limitations and responsible interpretation

• All observations and labels are synthetic; no patient data were used.

• This is not a cancer diagnostic model and does not establish clinical applicability.

• A mean AUC close to 0.5 is consistent with chance-level ranking in this experiment, but does not prove the true AUC is exactly 0.5.

• Observed values depend on sample size, number of features, model, feature-selection method, cross-validation design, and random seeds.

• For a formal research presentation, rerun the notebook, preserve the exact code and software versions, and consider uncertainty estimates and a separate untouched test set.

6. Conclusion

This experiment shows that evaluation design matters: a high score is not trustworthy if information from validation data influenced model development. In biomedical AI, preventing data leakage is essential before interpreting performance claims.

Prepared from the user's reported experimental outputs. Rerun the notebook to independently verify and reproduce the values before formal submission.